

Dexmedetomidine Sedation

Standard Operating Procedure for

Awake Fibreoptic Intubation

St George's Anaesthetic Department

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1. Background

Follow DAS guidelines on oxygenation, topicalization and performance of awake tracheal intubation. SOP for dexmedetomidine sedation is described here.

1.1 Evidence for Dexmedetomidine sedation for awake tracheal intubation

- 2014 Cochrane review¹
 - o vs Propofol TCI
 - Lower levels of procedural discomfort
 - Similar intubation times
 - Less airway obstruction
 - No significant increase in cardiovascular adverse effects when compared with fentanyl, midazolam or propofol-based techniques
 - o vs Remifentanyl
 - Higher rates of recall
 - Higher oxygen saturations

1.2 Contraindications

- Known hypersensitivity to dexmedetomidine

1.3 Precautions

- Advanced heart block, bradycardia
- Significant cardiac dysfunction (e.g. ventricular dysfunction, congestive cardiac failure, pulmonary hypertension) and patients in whom sympathetic tone is critical for maintaining hemodynamic balance
- Desensitised autonomic nervous system control e.g.: elderly, diabetics
- Hypovolemia

2. Dexmedetomidine sedation setup

2.1 Monitoring

- Secure peripheral IV access
- Continuous ECG monitoring
- Non-invasive blood pressure
- Continuous pulse oximetry

2.2 Drug presentation

- Glass ampoule – 200mcg/2ml (concentration 100mcg/ml)

2.3 Infusion preparation and administration

1. Dilute Dexmedetomidine 200mcg in 50ml 0.9% Sodium Chloride. ([Appendix 4.1](#) for drug appearance)

Final concentration 4mcg/ml

2. Weight-based dosing
 - a. BMI <30 - use patient's actual body weight (ABW)
 - b. BMI >30 – use patient's adjusted body weight (AdjBW)
([Appendix 4.4](#) for body weight calculations)
3. Set up infusion pump to give appropriate weight-based loading dose followed by infusion. Agilia pumps allow this to be programmed uninterrupted.

Recommended regime:

Loading dose – 1mcg/kg over 10 minutes

Infusion dose – 0.2-0.7 mcg/kg/h titrated to effect

([Appendix 4.2](#) for step-by-step guide using Agilia pump)

([Appendix 4.3](#) for manual dosing when Agilia pump not available)

4. Commence Dexmedetomidine infusion as you commence topicalization as per DAS ATI guidelines².
 - Note Dexmedetomidine may take more than 10 minutes to have an effect
5. Proceed with ATI, titrating maintenance dose of Dexmedetomidine, aiming for 2 or 3 on the Ramsay Sedation Scale (RSS) (as suggested in Cochrane review)³
([Appendix 4.5](#) for RSS)

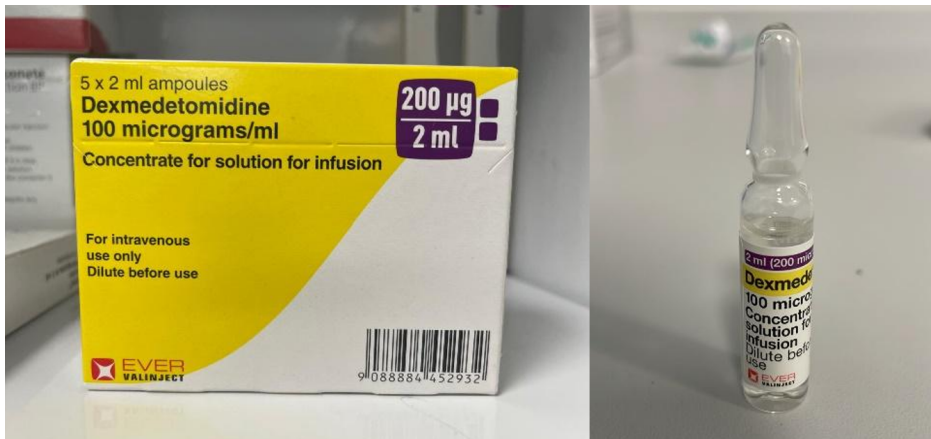
3. WFSA (World Federation of Societies of Anaesthesiologists) suggested dosing regimen for Dexmedetomidine AWFOI⁴

		Example for 60 kg adult	Notes
Step 1	Loading dose 1 mcg/kg Over 10-15 mins	60 mcg over 10-15 mins	This can be given whilst topically anaesthetising the airway
Step 2	Infusion 0.2-0.7 mcg/kg/hr	0.4 mcg/kg/hr (24mcg/hr)	Titrated to effect

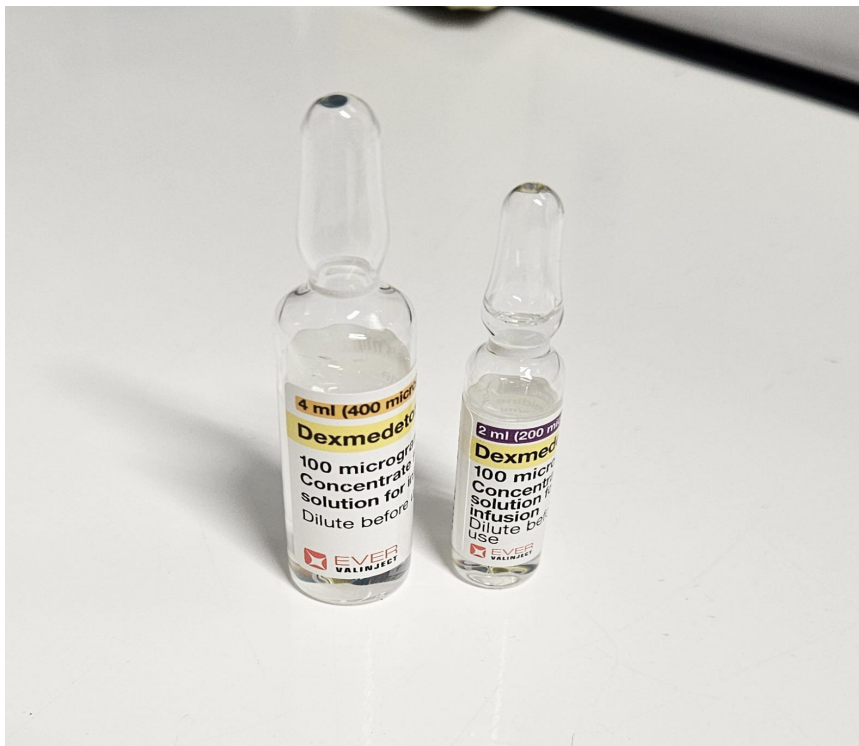
4. Appendices

Appendix 4.1 Dexmedetomidine ampoule appearance

200mcg/2 mL, 2mL vial (100mcg/mL)



Please be aware there is a 400mcg/4 mL vial (same concentration of 100mcg/mL)



[NB] Drug appearance subject to change depending on the pharmaceutical supply

Appendix 4.2 Agilia infusion pump set up pictorial guide

Step 1 – choose dose (not TCI)

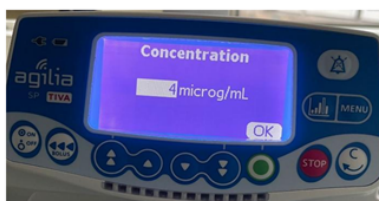


Step 2 – Concentration - choose mcg/m



Step 3 – input the dilution

Standard dilution of Dexmedetomidine is
200mcg/50ml = 4mcg/ml



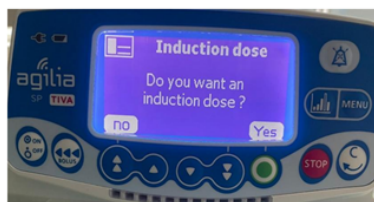
Step 4 – Choose infusion rate

This will be the maintenance rate that will
start after the loading dose.

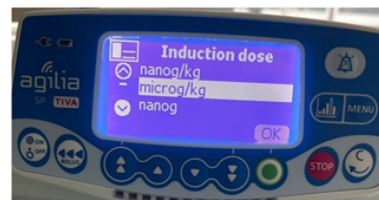


Step 5 – Induction dose

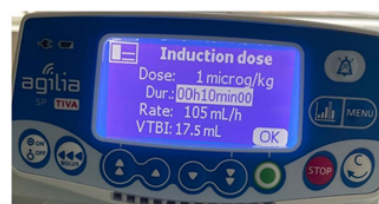
Select yes (induction dose = loading dose)



Choose mcg/kg



Choose induction dose and duration



(Usual induction dose 1mcg/kg over 10 mins)
Picture above:

- 1mcg/kg for 70kg adult
 - o The pump will deliver 17.5ml
 - o After the loading dose, the pump will automatically continue at the maintenance infusion rate.

*If using alternate infusion regimes, we suggest using an alarm to indicate end of bolus infusion to avoid overdosing.

Appendix 4.3 Dexmedetomidine dosing table based on weight

St. George's local clinical guideline.

Table can be used to calculate loading dose LD and infusion rates in absence of Agilia pump.

See below ml/h and VTBI (volume to be infused) for loading dose (LD) and maintenance.

Dexmedetomidine			Maintenance (ml/h)									
Loading dose (1 mcg/kg)	Loading (ml/h) for 10 min		Infusion rates with LD					without LD				
BW (kg)	ml/h for 10 min	VTBI (ml)	mcg/kg/h					mcg/kg/h				
			0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0	
50	75	13	3	4	5	6	8	9	10	11	13	
55	83	14	3	4	6	7	8	10	11	12	14	
60	90	15	3	5	6	8	9	11	12	14	15	
65	98	16	3	5	7	8	10	11	13	15	16	
70	105	18	4	5	7	9	11	12	14	16	18	
75	113	19	4	6	8	9	11	13	15	17	19	
80	120	20	4	6	8	10	12	14	16	18	20	
85	128	21	4	6	9	11	13	15	17	19	21	
90	135	23	5	7	9	11	14	16	18	20	23	
95	143	24	5	7	10	12	14	17	19	21	24	
100	150	25	5	8	10	13	15	18	20	23	25	
105	158	26	5	8	11	13	16	18	21	24	26	
110	165	28	6	8	11	14	17	19	22	25	28	
115	173	29	6	9	12	14	17	20	23	26	29	
120	180	30	6	9	12	15	18	21	24	27	30	

(Dosing table from IR conscious sedation for Renal cryoablation guideline – by Dr Ahmed Elwishi ⁵

Appendix 4.4 Calculations for weight-based dosing

- a. BMI <30 - use patient's actual body weight (ABW)

ABW = Actual body weight (kg)

- b. BMI >30 – use patient's adjusted body weight (AdjBW)

Need to calculate Ideal Body Weight (IBW) for Adjusted Body Weight (AdjBW)

IBW = Ideal body weight (kg)

Devine formula:

Males: $IBW = 50\text{kg} + 0.9\text{kg} \times (\text{height in cm} - 152\text{cm})$

Females: $IBW = 45.5\text{kg} + 0.9\text{kg} \times (\text{height in cm} - 152\text{cm})$

Example IBW for male 180cm tall and 100Kg actual weight (BMI 30.9)

$IBW = 50 + 0.9\text{kg} \times (180 - 152\text{cm}) = 50 + 0.9\text{kg} \times 28 = 75.2\text{Kg}$

AdjBW = Adjusted body weight (kg)

$AdjBW = IBW + 0.4 \times (ABW - IBW)$

$AdjBW = 75.2 + 0.4 \times (100 - 75.2) = 75.2 + 0.4 \times 24.8 = 75.2 + 9.92 = 85.12\text{Kg}$.

Alternatively, the SOBA app or other websites e.g. MDCalc.com can be used.

Appendix 4.5 Ramsay Sedation Scale

The Ramsay Sedation Scale

Clinical Score	Patient Characteristics
1	Awake; agitated or restless or both
2	Awake; cooperative, oriented, and tranquil
3	Awake but responds to commands only
4	Asleep; brisk response to light glabellar tap or loud auditory stimulus
5	Asleep; sluggish response to light glabellar tap or loud auditory stimulus
6	Asleep; no response to glabellar tap or loud auditory stimulus

Adapted from Ramsay MA, Savage TM, Simpson B, et al: Controlled sedation with alphaxalone-alphadolone. *British Medical Journal* 22;2(5920):656–659. doi: 10.1136/bmj.2.5920.656

5. References:

1. He, X. Y., Cao, J. P., He, Q., & Shi, X. Y. (2014). *Dexmedetomidine for the management of awake fiberoptic intubation*. Cochrane Database of Systematic Reviews, 2014(1), CD009798. <https://doi.org/10.1002/14651858.CD009798.pub2>
2. Ahmad, I., El-Boghdady, K., Bhagrath, R., Hodzovic, I., McNarry, A. F., Mir, F., O'Sullivan, E. P., Patel, A., Stacey, M., & Vaughan, D. (2019). *Difficult Airway Society guidelines for awake tracheal intubation (ATI) in adults*. *Anaesthesia*, 74(4), 509-528. <https://doi.org/10.1111/anae.14904>
3. Ramsay MA, Savage TM, Simpson B, et al: Controlled sedation with alphaxalone-alphadolone. *British Medical Journal* 22;2(5920):656–659. doi: 10.1136/bmj.2.5920.656
4. Crowe, G., Atterton, B., & Moran, L. (2022). *Perioperative applications of dexmedetomidine*. World Federation of Societies of Anaesthesiologists (WFSA) Virtual Library. <https://resources.wfsahq.org>
5. Elwishi, A. *IR Conscious Sedation for Renal Cryoablation Guideline*. St George's Hospital.